

Timeline Development

PERT Estimation Technique

Each Level 3 work package from the Work Breakdown Structure is estimated using the Program Evaluation Review Technique (PERT). Three time estimates are gathered per task — Optimistic (O), Most Likely (M), and Pessimistic (P), each in working days — and combined into a single Expected Time (Te) using the weighted formula:

$$Te = (O + 4M + P) / 6$$

This weighs the most likely estimate four times more heavily than the best- and worst-case scenarios, producing a realistic single duration for planning purposes.

PERT Estimation Table (Level 3 Work Packages)

Code	Work Package	O	M	P	Te = (O+4M+P)/6
1.0	Project Documentation				Σ 8.34 days
1.1	Project Proposal	2	3	5	3.17 days
1.2	WBS & Timeline	1	2	3	2.00 days
1.3	User Story Document	1	1.5	2	1.50 days
1.4	Use Case Diagram	1	1.5	3	1.67 days
1.5	Sprint Planning & Progress Update (per sprint)	0.5	1	2	1.08 days
1.6	Sprint Retrospective (per sprint)	0.5	1	1.5	1.00 days
1.7	Stand-up Coordination (per week)	0.5	0.5	1	0.58 days

Code	Work Package	O	M	P	Te = (O+4M+P)/6
2.0	UI/UX Design				Σ 13.17 days
2.1	Low-Fidelity Wireframes	2	3	5	3.17 days
2.2	High-Fidelity Mockups	3	5	7	5.00 days
2.3	Style Guide	1	2	3	2.00 days
2.4	Responsive Layout Specs	2	3	4	3.00 days
3.0	User Authentication				Σ 10.17 days
3.1	Sign-Up UI	2	3	4	3.00 days
3.2	Account Creation API	1	2	3	2.00 days
3.3	Login UI	1	2	3	2.00 days
3.4	Auth API & Session Handling	2	3	5	3.17 days
4.0	Project Mgmt. Feature				Σ 9.67 days
4.1	Create/Edit Project UI	2	3	5	3.17 days
4.2	Project API Endpoints	2	3	4	3.00 days
4.3	Project List/Switcher UI	1	2	3	2.00 days
4.4	Project Data Model	1	1.5	2	1.50 days
5.0	Task Mgmt. Feature				Σ 16.34 days
5.1	Create/Edit/Delete Task UI	3	4	6	4.17 days
5.2	Task API Endpoints	2	4	6	4.00 days

Code	Work Package	O	M	P	Te = (O+4M+P)/6
5.3	Due Date & Deadline Feature	1	2	3	2.00 days
5.4	Recurring Task Feature	2	3	5	3.17 days
5.5	Priority Level Feature	1	1.5	2	1.50 days
5.6	Labels/Tags Feature	1	1.5	2	1.50 days
6.0	Dashboard & Reporting				Σ 5.17 days
6.1	Current & Upcoming Tasks View	2	3	5	3.17 days
6.2	Overdue Tasks View	1	2	3	2.00 days
7.0	Backend Infrastructure				Σ 10.18 days
7.1	Database Schema Design	2	3	5	3.17 days
7.2	Database Deployment	1	1.5	3	1.67 days
7.3	API Specification Doc	2	3	5	3.17 days
7.4	API Testing	1	2	4	2.17 days
8.0	QA, Deployment & Presentation				Σ 10.17 days
8.1	Test Matrix	1	2	4	2.17 days
8.2	Bug/Defect Reports	1	2	4	2.17 days
8.3	Hosted Web Application	1	1.5	3	1.67 days

Code	Work Package	O	M	P	Te = (O+4M+P)/6
8.4	GitHub Repo & README	0.5	1	2	1.08 days
8.5	Presentation Slide Deck	1	2	3	2.00 days
8.6	Live Project Demo	0.5	1	2	1.08 days

Grand Total (one-time work packages, excluding recurring ceremonies): Σ 83.21 person-days of estimated effort. Recurring tasks (sprint planning, retrospectives, stand-ups) are estimated per occurrence; their totals above reflect a single instance, not the full-term sum, since they repeat across every sprint alongside development work rather than adding sequential duration.